# Respiratory Polygraphy

## Sleep Study

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# Your referral to Independent Physiological Diagnostics

## Quick Start Guide

- Introduction
- Setting up for your sleep study
  - Applying the Alice NightOne
  - During the night
  - In the morning
- Troubleshooting
- Warnings and Cautions

You have been referred by your clinician to Independent Physiological Diagnostics for a home sleep study.

The sleep study you have been referred for is a Respiratory Polygraphy and will be provided with a Philips Alice NightOne diagnostic device. This test procedure records respiratory parameters over the course of the night and is used as to investigate sleep disordered breathing (SDB), diagnosing conditions such as sleep apnoea. This guide has been developed to guide you through set-up and troubleshoot any problems you may have.

If you have any questions, feel free to reach out to our friendly team on (020) 7381 8698.

# Introduction

The **Alice NightOne** unit records physiological information while you sleep. The purpose of this study is to evaluate your breathing while you're sleeping.

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# The icons that will be available on your Alice NightOne device are as follows:

### Respiratory effort (chest) belt

The belt should **double click** and insert securely into the ANO device and recording starts as soon as the device is buckled. The belt is fitted around your chest and senses chest movement during breathing.

### Airflow sensor

The airflow sensor measures your breathing throughout the night. The prongs are placed or inserted into the tips of your nostrils. The airflow sensor has a clear twist lock connector and is connected at the top of the Alice NightOne device

### SpO2 sensor

The SpO2 sensor measures the amount of oxygen in your blood and your pulse rate. The SpO2 sensor attaches to your forefinger on your non-dominant hand

### Good Study Indicator (GSI)

The Good Study Indicator will show how much time was recorded and whether the quality of signals were good. It will also indicate any problem sensor(s)

## Smart Guide Patient Setup

**Step 1.** Effort (chest) belt connection icon

**Step 2.** Cannula connection icon

**Step 3.** SpO2 connection icon

**Step 4.** Good Study Indicator icon

# Setting up for your Sleep Study

We recommend that you wear a vest or t-shirt for your Sleep Study. Please do not fit the sensors directly onto the skin.

# Applying the Alice NightOne

You can watch a video online to guide you through the set up at https://youtu.be/-vaXa6ECIUU

**1. Connect the effort (chest) belt**

- Buckle the Alice NightOne around your chest

The two red dots on the connectors should be facing outwards.

The belt will need to be double clicked into place securely.

For Men: Place the belt around the body just below the nipple line, between your chest and abdomen

For Women: Place the belt around the body

- Alice NightOne will automatically power on (please wait while the device takes a moment to start up)
- If the chest belt icon flashes yellow, tighten the belt around you, using the adjustment straps
- Do not overtighten the belt – the fit should be snug.

(See troubleshooting for belt adjustment tips)

- The belt icon should now be solid green

**2. Connect the airflow sensor**

- Connect the airflow sensor to Alice NightOne (if not already connected)
- Loop the airflow sensor around your ears and under your chin
- Adjust the slider to fit comfortably under your chin
- Tape the airflow sensor to your cheeks, using the tape provided
- The airflow sensor icon should be solid green

**3. Connect the finger probe (SpO2 sensor)**

- Connect the finger probe to the index or second finger of your non-dominant hand.

Please remove any nail varnish or false/acrylic nails (if worn)before connecting the finger probe

- The cable will rest on top of your finger and on the back of your hand
- Tape the wire to your hand, using the tape provided.

Do not tape over the probe itself.

-The finger probe icon should now be green

The finger probe you have been given may differ from the image

Once all of the sensors have been applied, the lights will turn to solid green

Over the next few minutes all the lights will turn off gradually one by one

**4. You can now go to bed and fall asleep**

# During the night

If the device notices a problem with a sensor, a flashing yellow icon will appear for ten minutes.

If you notice this, please adjust the sensor that is indicated. The flashing icon will turn from yellow to green and then disappear.

You can also check the sensor status at any time by briefly pressing the user button on the front of the device.

**- Solid green lights indicate that the sensors are connected and working properly**

**- If a light is flashing yellow, adjust the sensor that is indicated (see troubleshooting below)**

# In the morning

In the morning, once the study is complete and you are ready to wake up for the day, remove the sensors from your body and disconnect the Alice NightOne device.

**- Press and hold the user button for five seconds to turn the device off**
**- Otherwise, the device will turn itself off after 30 minutes**

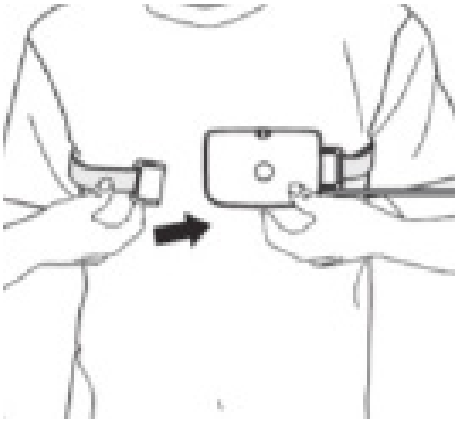
Place the Alice NightOne and all sensors into the carrying case, along with this quick start guide.

# Troubleshooting

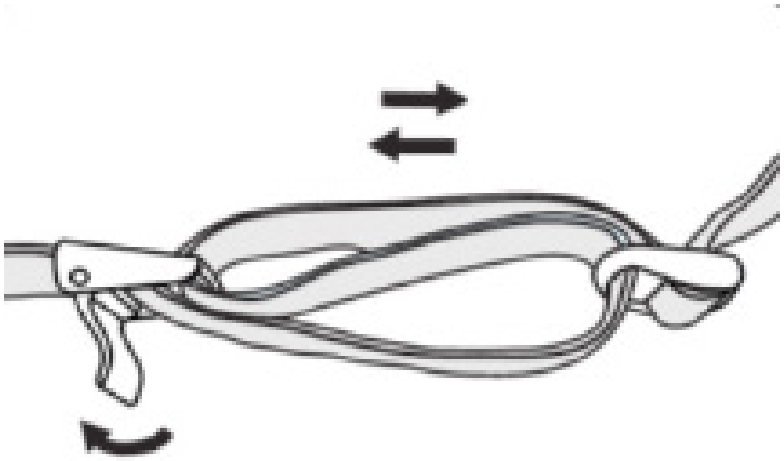
## 1. Fitting the belt

### Adjustment tips:

- Adjustment to the belt must be done whilst you are sat upright
- To adjust, move the slide lock away from the buckle to tighten; move slide lock closer to loosen
- There should be no slack in the three layers of belt between the clasp and slide lock



**Effort Belt Connection**



**Effort Belt Adjustment**

## 2. Smart guide patient set up

During setup, illuminated sensor indicators will walk you through sensor placement.

### The smart guide gives you feedback that:

- Each sensor is connected properly
- The signal quality is good
- The device is functioning properly

| Problem | Solution |
| --- | --- |
| **SpO2 sensor icon continues to flash yellow** | Make sure the finger probe is correctly and securely placed on your finger. If the sensor icon continues to flash after adjusting the sensor, you should go to sleep as planned. |
| **Effort belt sensor icon continues to flash yellow** | Make sure the effort belt is securely connected to the Alice NightOne device. Also, make sure the belt is placed correctly around your chest and fits snugly to ensure there is enough tension on the belt. If the belt is too tight, slightly loosen the belt to ensure there is not too much tension. If the sensor icon continues to flash after adjusting the sensor, you should go to sleep as planned. |
| **Airflow sensor icon continues to flash yellow** | Make sure the airflow sensor is properly routed behind your ears and under the chin with the airflow sensor prongs facing up inside your nostrils. Also, make sure you are breathing through your nose. If the sensor icon continues to flash after adjusting the sensor, you should go to sleep as planned. |
| **Effort belt buckle unsnaps from the device** | Make sure both buckle latches are firmly seated into place when connecting to the device. |
| **Alice NightOne will not start the study** | Press and hold the user button or connect the effort belt to initiate the guided setup and begin recording. If no lights are displayed, remove and replace the batteries and try again. |
| **When trying to display the Good Study Indicator at the end of a study, the guided setup begins** | Press and hold the user button until the Good Study Indicator flashes. Then, quickly press and release the user button and the Good Study Indicator will be displayed. |
| **Two sections (the first quadrant and the fourth quadrant) of the Good Study Indicator are flashing yellow, no sensor icons are displayed, and the device does not respond to any button presses** | There may be a device error. |

## Warnings and Cautions

- Do not smoke when using the device
- Use Micropore tape to secure the SpO2 cable to the back of the hand and also to secure airflow sensors to cheeks.
- To avoid sensor damage, do not immerse any of the sensors in liquid.
- Remove the airflow sensors from your face before drinking liquids or eating.
- To avoid electric shock, do not wear the Alice NightOne and its sensors in a shower, bath, waterbed, or with an electric blanket that is switched on.
- Use only the sensors specified by Philips Respironics for the Alice NightOne unit. Non-approved Philips Respironics sensors may compromise the safety and performance of the device.
- Do not use a mobile phone near the Alice NightOne recorder when it is powered on. The radio frequencies from the mobile may interfere with the operation of the device.

## Next steps

Your referring doctor will receive analysed sleep physiologist and sleep physician report within 5 working days. You may have already booked your follow up appointment to discuss the findings. The next steps after a sleep study will depend on the specific findings of the study and the underlying sleep disorder or medical condition you are being evaluated for. Your doctor or sleep specialist will provide you with guidance and recommendations based on the results of your study. Some of the possible scenarios may be:

1. Regardless of the results of the sleep study, you may be advised to make lifestyle changes such as losing weight, quitting smoking, and avoiding alcohol and caffeine before bedtime to improve your overall sleep quality.
2. If the sleep study shows no significant abnormalities or sleep disorder, then no further treatment may be necessary, and you may simply continue with your normal sleep habits.
3. In rare cases, you may be referred for a repeat and more sensitive sleep study such as Home Video Polysomnography.
4. In some cases, if the sleep study reveals a sleep disorder that requires intervention, such as sleep apnea you may be referred for therapy such as Continuous positive airway pressure (CPAP), Mandibular Advancement Device (MAD) or Non-invasive ventilation (NIV).

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